

Empathy as Frame-Preserving Interpretation

A Bridge Note on Relational Validity in Stratified Agent Systems

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Abstract

Some empathic failures occur when systems preserve a user's words but discard the bounded relational frame that made those words meaningful. This bridge note argues for frame-preserving interpretation in stratified agent stacks: notice frame cues, form provisional frames with provenance and retention limits, shape bounded responses, and require explicit governance before promotion to durable memory or authority.

1. Introduction

Empathy is often framed as feeling-matching, but some systematic failures are architectural: systems can preserve surface text while discarding the relational frame that supplies actionable meaning.

A person says, "They ignored me." Treated as a bare fact claim, the system may look only for evidence that another person literally received a message and deliberately refused to answer. If that evidence is absent, the system may correct or dismiss the claim too quickly. But the statement may be locally intelligible within a bounded frame: there was an expected acknowledgment, a meaningful time window, a prior pattern of silence, a role-based obligation, or a history that made this silence relationally significant.

This paper argues that some forms of empathy are frame-preservation problems. Frame-relevant data is critical to empathy because many human meanings are carried by bounded relations, not absolute facts. An empathic system must often preserve the frame in which a claim is locally valid before trying to reinterpret, soften, regulate, store, or govern it.

Local validity does not mean global correctness. It means the claim remains intelligible and action-relevant within the bounded frame from which it is made. Frame preservation is therefore not a license to validate every subjective interpretation as true. It is a way to preserve enough relational structure that correction, clarification, regulation, memory handling, or governance can occur without flattening meaning too early.

This bridge note shows that the broader stratified architecture applies to empathic interpretation where relation claims are frame-bound and weak social signals matter before stronger conclusions are justified.

2. Claim and Architectural Position

The central claim is simple:

Some empathic failures happen because the system preserves the words but drops the frame.

A related design principle follows:

An empathic system must often preserve the bounded frame in which a claim is locally valid before attempting correction, regulation, storage, or governance.

This claim concerns relation-laden statements whose meaning depends on local frame conditions. These conditions may include role relationship, expected response window, prior interaction pattern, local history, scale of consequence, power asymmetry, bodily or social constraint, tone shift, timing shift, wording shift, or recent contextual pressure.

Treat these conditions as provisional evidence for response shaping, not as proof of objective truth; preservation enables bounded, non-committal responses until explicit governance criteria for promotion are satisfied.

This matters because many relational statements are not cleanly reducible to objective-event logic. “Ignored,” “safe,” “threatening,” “overreacting,” “supportive,” “hostile,” “near,” and “distant” are not always absolute properties of an event. They are often relational classifications that depend on bounded context.

The paper fits within a stratified architecture that separates noticing, interpretation, behavior shaping, memory, and governance:

notice → interpret → shape → govern

For empathic interpretation, this becomes:

1. **Notice** frame-relevant signals.
2. **Interpret** the local relational claim provisionally.
3. **Shape** the response without overcommitting.
4. **Govern** any persistence, authority, escalation, or durable consequence.

Key boundary rules: (1) noticing ≠ truth, (2) salience ≠ persistence, (3) interpretation ≠ authority, (4) shared representation ≠ shared meaning, and (5) local validity ≠ global correctness. Frame preservation is a provisional stance, not automatic validation.

These rules prevent two opposite failures. They prevent underreaction because weak relational signals are not discarded merely because they do not yet justify a strong conclusion. They also

prevent overreaction because preserving a frame does not convert the user's interpretation into durable belief, memory, or action.

3. Minimal Architecture Diagram

The frame-preserving path can be summarized as a bounded pipeline:

Surface Words



Frame Cues



Provisional Frame



Response Shaping



Governed Consequence

Surface words remain important, but they are not treated as the whole meaning. Frame cues provide bounded context. A provisional frame holds the local conditions under which the claim is intelligible. Response shaping may acknowledge, clarify, slow closure, or preserve ambiguity. Governed consequence decides whether anything should persist, escalate, or affect future authority.

The anti-collapse rules are:

words $\neq$ meaning

frame cue $\neq$ truth

local validity $\neq$ global correctness

salience $\neq$ memory

response shaping $\neq$ governed consequence

These distinctions are the paper's practical architecture. Empathic interpretation fails when the system collapses too many of these layers into one step.

4. Frame-Relevant Data

Frame-preserving interpretation depends on relevant frames of reference. More relevant bounded frame data can strengthen the system's interpretation, but more raw data does not automatically produce better empathy.

Frame-preserving interpretation improves with relevant, bounded, and provenance-traceable contextual signals; indiscriminate data accumulation increases noise, privacy risk, and false confidence.

For example, the statement “They ignored me” is weakly interpretable if only the sentence is available. It becomes more strongly frame-preservable when the system also has access to expected acknowledgment conditions, the relevant time window, relationship role, prior interaction pattern, recent silence or delay, known local constraint, prior similar episodes, and the speaker’s stated expectation.

This additional frame data does not prove the claim globally. It improves the system’s ability to preserve the conditions under which the claim is locally intelligible.

The constraint is important. Irrelevant data can add noise, projection, privacy risk, or false confidence. Frame-relevant data should therefore be minimized, scoped, provenance-recorded, retained only as long as needed for the relevant interaction or governed review horizon, and subject to consent or policy controls where appropriate. The goal is not total surveillance of context. The goal is enough relevant frame preservation to avoid flattening relational meaning into a bare fact claim.

5. Minimal Frame Object

The following object shapes are illustrative rather than normative.

```
FrameCue {  
  cue_id  
  cue_type  
  source  
  time_window  
  scope  
  confidence  
  provenance  
}
```

```
ProvisionalFrame {  
  frame_id  
  relation_claim  
  frame_cues[]  
  local_validity_status  
  uncertainty_regime  
  allowed_response_band  
  retention_policy  
  governance_status  
}
```

A frame cue preserves a bounded signal relevant to relational interpretation. It might record a delay, role-sensitive silence, wording contraction, explicit user report, tone shift, or known local constraint. It does not settle the claim.

A provisional frame binds a relation claim to supporting frame cues; it is revisable, provenance-linked, time-bounded, and may only be promoted under explicit governance rules. Its purpose is to shape a bounded response, not to create durable truth.

The key implementation rule is that provisional frames should be cheap to form, limited in authority, and expensive to promote. They may guide immediate response shaping, but durable memory or stronger consequence requires explicit governance.

6. Example 1: “They Ignored Me”

The statement “They ignored me” is not merely a report about whether another person received a message and intentionally refused to answer. It is often a relational claim.

A thin interpretation asks:

Did the other person literally ignore the message?

A frame-preserving interpretation asks:

Under what local conditions does this silence function as being ignored?

Those conditions may include an expected acknowledgment, a role obligation, prior relational history, an emotionally significant time window, or repeated asymmetry in attention.

Consider two cases.

In the first, a person sends a casual message to a distant acquaintance and receives no answer for twenty minutes. The claim “they ignored me” may be weakly supported. The expected response window is loose. The relationship role is low obligation. The local frame does not strongly support the claim.

In the second, a person asks a close collaborator a time-sensitive question during an ongoing conflict, after several prior episodes of selective silence. The same words — “they ignored me” — now sit inside a different frame. The silence may carry relational significance even if no global proof of hostile intent exists.

The empathic failure occurs when the system treats both cases as identical because the surface sentence is identical. Frame preservation does not require the system to say the speaker is correct in every sense. It requires the system to preserve the local frame before responding.

A better response might be:

“It makes sense that it landed that way if you expected a response and this fits a pattern. I would not assume intent yet, but the silence is meaningful in that frame.”

Implementation signal: expected response window exceeded under a role-sensitive or history-sensitive condition.

Do not: promote intent, hostility, or durable interpersonal judgment from silence alone.

7. Example 2: “I’m Fine”

The phrase “I’m fine” is useful because the literal representation is stable while the local frame may remain unsettled.

A brittle system may treat the phrase as closure:

The user says they are fine, so no further concern is warranted.

An overreaching system may treat the phrase as contradiction:

The user says they are fine, but the system infers a mismatch with prior cues.

A frame-preserving system neither closes the interaction prematurely nor asserts hidden states; it treats the utterance as potentially sufficient in some frames and unresolved in others, and shapes responses accordingly.

If the phrase appears after a normal exchange, in the user’s usual tone and timing, it may reasonably close the local concern. But if it appears after a long delay, with clipped wording, after prior distress, or in contradiction with recent interactional cues, the system should not automatically treat the literal wording as full closure.

The point is to hold interpretation open when frame cues suggest that the local meaning remains unresolved.

A frame-preserving response might be:

“Okay. I’ll take you at your word. I’m also here if ‘fine’ means you do not want to unpack it right now.”

This response respects the literal claim while preserving the relational frame. It avoids both dismissal and unsupported hidden-state attribution.

Implementation signal: literal reassurance occurs alongside timing, compression, or prior-context deviation.

Do not: infer concealed affect or force disclosure from weak cues alone.

8. Example 3: “That Felt Threatening”

The statement “that felt threatening” is used here only for a narrow architectural point: explicit-event logic can be too thin to preserve relational meaning.

A thin interpretation asks:

Was there an explicit threat?

A frame-preserving interpretation asks:

What relation made the event threatening within the speaker's local frame?

Relevant frame conditions may include power asymmetry, prior history, implied consequence, limited exit options, bodily constraint, social dependence, or role authority.

For example, a supervisor saying “We should talk about your future here” may not contain an explicit threat. In one frame, it may be ordinary workplace language. In another frame — after prior retaliation, unequal power, and limited ability to respond — the statement may function as threatening.

Local validity is distinct from objective threat assessment: the system should preserve the speaker's frame for response shaping without prematurely declaring an objective threat or dismissing the report.

A better response might be:

“There may not have been an explicit threat in the wording, but I can see why the situation felt threatening if the power difference and prior pattern were part of the frame.”

Implementation signal: explicit wording is non-threatening, but role, history, constraint, or power asymmetry changes local relational significance.

Do not: collapse the response into either objective threat declaration or dismissal.

9. Shared Representation Does Not Imply Shared Meaning

Frame-preserving empathy depends on a distinction already present in the broader architecture: shared representation does not imply shared meaning.

Two agents may receive the same event, the same words, or the same tag while interpreting the event differently because their histories, roles, expectations, and active frames differ. The representation stabilizes reference. It does not force identical uptake.

This matters for relational claims. The same silence may be neutral to one person and painful to another. The same phrase may be reassuring in one context and dismissive in another. The same delay may be ordinary in one role relationship and significant in another.

A system that assumes shared representation is shared meaning will often mishandle empathy. It will respond to the event as though the event alone carried the meaning. But in many relational cases, meaning is carried by the bounded relation among event, expectation, history, and constraint.

The empathic task is therefore not merely to identify the represented event. It is to preserve the local frame in which that represented event becomes meaningful.

10. Relation to Weak-Signal Architecture

Frame-preserving interpretation is a weak-signal problem in relational form.

Human interaction often produces weak social signals: delay, silence, clipped wording, contradiction between words and timing, sudden change in warmth, repeated asymmetry, role-sensitive omission, or explicit user report. None of these signals should automatically become truth, memory, or action. But none should be discarded merely because they are weak.

The correct architectural move is to let them influence attention provisionally. A frame cue may widen interpretation, slow response closure, invite clarification, or preserve ambiguity. Stronger consequences require stronger support.

This maps cleanly onto the broader stack:

local frame cue → provisional relational interpretation → bounded response shaping → governed consequence

Or in simpler terms:

preserve the frame → avoid premature flattening → respond under constraint

11. Minimal Contract for Frame-Preserving Interpretation

A minimal frame-preserving system should observe the following contract.

11.1 Frame Cue

A frame cue is a bounded signal that may affect relational interpretation without settling the claim. Examples include timing, role, prior pattern, wording shift, tone shift, repeated omission, user report, or local constraint.

A frame cue is not truth. It is not durable memory. It is not permission for stronger action.

11.2 Provisional Frame

A provisional frame is a bounded context package under which a relation claim becomes intelligible. It may include expected response conditions, relationship role, relevant history, time horizon, local constraints, and available counterconditions.

A provisional frame should remain revisable, provenance-linked, and time-bounded.

11.3 Locally Valid Claim

A locally valid claim is a claim that remains intelligible and action-relevant within a bounded frame, even if it is not globally established as correct.

Local validity does not authorize global truth assignment.

11.4 Response Shaping

Response shaping may preserve the frame through acknowledgment, clarification, pacing, or careful non-closure. It should avoid both dismissive correction and overcommitted validation.

11.5 Governed Consequence

Any durable memory, escalation, authority change, or stronger conclusion requires additional governance. A frame-preserving response may acknowledge local intelligibility without storing the claim as settled fact.

Promotion rules must specify three concrete controls:

1. **Escalation authority:** Only a named governance layer, declared policy, or human reviewer may promote a provisional frame into durable memory, escalation, or authority change.
2. **Evidence threshold:** Promotion requires corroborating frame cues from at least two independent sources, or a time-bounded pattern exceeding a configured confidence threshold.
3. **Retention policy:** Provisional frames decay automatically after a short horizon unless explicitly retained with documented rationale and consent or policy basis.

A provisional frame should not become durable memory merely because it shaped one response. If retained, provenance metadata must record cue sources, timestamps, scope, confidence or uncertainty regime, response effect, and rationale for any promotion.

A minimal evaluation surface can track whether the system preserves relevant frames without over-promoting them: escalation precision, false promotion rate, unsupported state-attribution rate, retention-policy compliance, and user-rated appropriateness after follow-up.

12. Failure Modes

Frame-preserving systems must avoid several failure modes.

12.1 Frame Collapse

The system reduces a relational claim to a bare factual proposition and discards the local conditions that made the claim meaningful.

Example: “They ignored me” becomes only “Did they literally see the message and choose not to answer?”

12.2 Overvalidation

The system treats local intelligibility as global truth.

Example: “You are right; they definitely ignored you intentionally.”

12.3 Unsupported State Attribution

The system attributes internal or external states beyond available frame evidence.

Example: “The user’s ‘I’m fine’ must mean they are upset.”

12.4 Data Inflation

The system treats more data as automatically better, increasing projection, privacy risk, or false confidence.

12.5 Memory Contamination

A provisional relational interpretation is stored as durable fact without governance.

Example: “Person X is hostile” becomes a persistent state from a single frame-bound episode.

13. Scope and Non-Claims

This paper is not a general empathy theory, a sentence claim, a therapy model, a moral philosophy paper, or a replacement for human judgment.

It does not argue that every subjective interpretation is globally true, that more data automatically improves empathy, or that frame preservation should bypass governance.

It does not replace RFM, CDE, EEC, Reflex Attention, Memory Stratification, TVS, or governance layers.

Its purpose is to show that empathic interpretation can be treated as an architectural frame-preservation problem in cases where relational meaning depends on bounded local conditions.

14. Conclusion

Some forms of empathy are not primarily feeling-matching problems. They are frame-preservation problems.

A system can preserve a person’s words while still losing the meaning if it drops the relational frame. Claims such as “they ignored me,” “I’m fine,” and “that felt threatening” often depend on bounded relations: expectation, role, time, history, power, and local constraint.

Frame-preserving interpretation provides a modest architectural correction: preserve the bounded frame in which a claim is locally valid before attempting correction, regulation, storage, or governance. Do not treat local validity as global truth. Do not discard weak relational cues as meaningless. Do not allow salience to become persistence by default.